CHAPTER 24 · DERIVATIONS

WAVE OPTICS

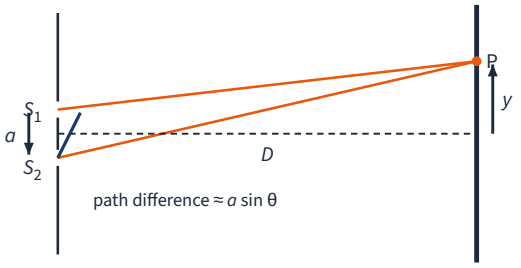
4 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 FRINGE SEPARATION IN YOUNG'S EXPERIMENT

SYLLABUS 24.3(B)

RESULT $x = \frac{\lambda D}{a}$

1. Slits S_1 and S_2 a apart; screen at distance D ($D \gg a$). Point P is at height y above the centre.
2. Path difference $S_2P - S_1P \approx a \sin\theta \approx a \tan\theta = \frac{ay}{D}$ (small θ).
3. Bright fringe: $\frac{ay}{D} = m\lambda$, so $y_m = \frac{m\lambda D}{a}$.
4. Fringe separation $\mathbf{x = y_{m+1} - y_m = \frac{\lambda D}{a}}$.

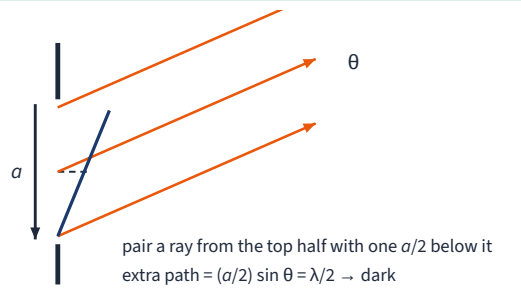


2 FIRST MINIMUM OF A SINGLE SLIT

SYLLABUS 24.5(B)

RESULT $\sin\theta = \frac{\lambda}{a}$

1. Divide the slit of width a into two halves.
2. Every point in the top half pairs with a point $\frac{a}{2}$ below it. In direction θ the path difference of each pair is $\frac{a}{2}\sin\theta$.
3. If $\frac{a}{2}\sin\theta = \frac{\lambda}{2}$, each pair cancels and the screen is dark.
4. $\mathbf{\sin\theta = \frac{\lambda}{a}}$ for the first minimum.



3 DIFFRACTION GRATING

SYLLABUS 24.6(B)

RESULT $d \sin\theta = m\lambda$

1. Grating spacing $d = \frac{1}{N}$ (N lines per metre).
2. In direction θ, light from adjacent slits has path difference d sin θ.
3. Bright maxima when all slits are in phase: $\mathbf{d \sin\theta = m\lambda}$ (m = 0, 1, 2, …), with m less than $\frac{d}{\lambda}$.

4 BREWSTER'S LAW

SYLLABUS 24.7(F)

RESULT $\tan \theta_B = \frac{n_2}{n_1}$

1. At the Brewster angle the reflected light is completely plane polarised and the reflected and refracted rays are at 90°.
2. $\theta_B + \theta_r = 90°$.
3. Snell's law: $n_1 \sin \theta_B = n_2 \sin \theta_r = n_2 \sin(90° - \theta_B) = n_2 \cos \theta_B$.
4. $\mathbf{\tan \theta_B = \frac{n_2}{n_1}}$.

